Chapter 2

Four Important Linear Partial Differential Equations

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 2; printed pp. 17–87; PDF pages 34–104.

Chinese reference: 中译本：书内第 13 页起；球均值见第 18–19 页，热球均值见第 35–37 页。

Editorial study edition. The exposition and proofs are reconstructed in the order of Evans, Chapter 2; they are not a verbatim transcription. Source theorem numbers are identified separately from our chapter-wide numbering. Chinese mathematical notes, worked examples, figures, and exercise solutions are editorial additions.

Our main route is

$$\text{fundamental solutions} \longrightarrow \text{representation formulas} \longrightarrow \text{estimates and uniqueness.}$$

The central comparison is between harmonic means (§2.2.2) and the heat-ball mean (§2.3.2). The latter is a positive weighted average over past space-time, not a uniform average over an ordinary ball.

四个模型四种机制

传输沿特征线搬运；Laplace 方程描述静态平衡；热方程耗散并向未来扩散；波动方程保存能量并有限速传播。相同的“平均”符号背后可能是完全不同的测度。

Conventions and prerequisites. $\omega_n = |B_1|$ and $|\partial B_1| = n\omega_n$. The elliptic operator is $-\Delta$; the heat operator is $L = \partial_t - \Delta_x$; the wave speed is one unless stated otherwise. $C^{2,1}$ means two continuous spatial derivatives and one continuous time derivative. Basic multivariable calculus, Green's identities, polar integration, dominated convergence, convolution, and smooth cutoffs are prerequisites. Boundary correctors on general domains are explicitly conditional on their existence; ball and half-space kernels are constructed here.

2.1 Transport Equation

2.1.1 Characteristics and the initial-value problem (Evans §2.1.1)

For fixed $b \in \mathbb{R}^n$, consider $u_t + b \cdot Du = 0$, $u(x,0) = g(x)$. Through (x,t) trace the line

$$X(s) = x + (s-t)b, \qquad 0 \le s \le t.$$

For a C^1 solution, the chain rule gives

$$\frac{d}{ds} u(X(s),s) = Du(X(s),s) \cdot b + u_t(X(s),s) = 0.$$

Integrating from 0 to t proves necessity of

$$\boxed{u(x,t) = g(x-tb).}$$

Conversely, if $g \in C^1$, differentiation of this formula yields $u_t = -b \cdot Dg(x-tb)$ and $Du = Dg(x-tb)$, proving the PDE and initial condition. This establishes both existence and uniqueness in the classical class, rather than merely guessing a candidate.

回溯特征线

计算当前点 (x,t) 的值，要沿速度 b 回到初始平面，落点是 $x-tb$。正负号可用一维脉冲向右移动检验：$g(x-bt)$ 的峰位于 $x = bt$。

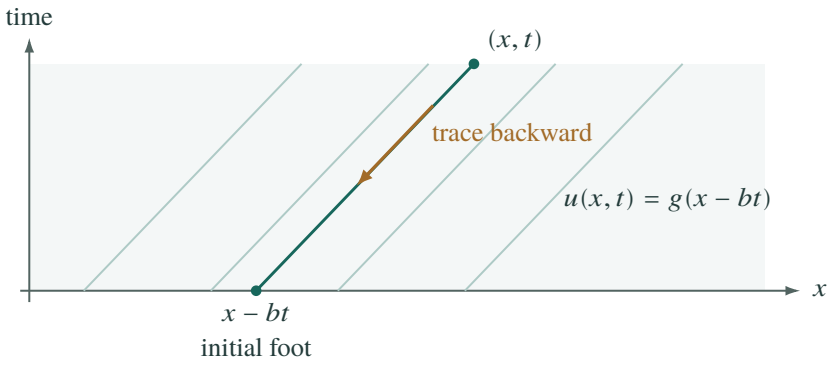


Diagram T2.1. Read a transport solution backward. The solution is constant on each characteristic when the source is zero. With $b > 0$, the forward line tilts right, while evaluation at a target requires its leftward initial foot $x - bt$. The orange arrow indicates evaluation backward in time, not a negative physical velocity.

Nonsmooth data and weak solutions. For $g \in L^1_{\mathrm{loc}}$, the same formula is locally integrable. For a compactly supported smooth test function on $\mathbb{R}^n \times [0,\infty)$, change variables $x = z + tb$:

$$\begin{aligned}\int_0^\infty \int u(\phi_t + b \cdot D\phi)\,dx\,dt &= \int_{\mathbb{R}^n} g(z) \int_0^\infty \frac{d}{dt}\phi(z+tb,t)\,dt\,dz \\ &= -\int_{\mathbb{R}^n} g(z)\phi(z,0)\,dz.\end{aligned}$$

Fubini is justified by compact test support and local integrability. This is the distributional formulation with the initial datum included. There is no need to assign a classical derivative at a moving jump. Translations preserve L^p norms: $\|u(\cdot,t)\|_p = \|g\|_p$ whenever $g \in L^p$.

2.1.2 Forcing and Duhamel's principle (Evans §2.1.2)

If $u_t + b \cdot Du = f$ and $u(x,0) = g(x)$, then

$$\frac{d}{ds}u(X(s),s) = f(X(s),s).$$

Integration along the same characteristic gives

$$\boxed{u(x,t) = g(x-tb) + \int_0^t f(x-(t-s)b,s)\,ds.}$$

For C^1 data differentiate under the integral, with the upper endpoint contributing $f(x,t)$; the remaining integrals cancel between u_t and $b \cdot Du$. This directly verifies the formula.

For the additional constant term cu, multiply along characteristics by e^{cs}:

$$u(x,t) = e^{-ct}g(x-tb) + \int_0^t e^{-c(t-s)}f(x-(t-s)b,s)\,ds.$$

The damping factor belongs to the elapsed time $t-s$, not to s alone.

源项沿轨迹累积

在时间 s 注入的源，之后还要平移 $(t-s)b$ 才到达当前点。Duhamel 的共同思想是把每个时刻的小冲量，按齐次方程传播后相加。

2.2 Laplace's Equation

2.2.1 Fundamental solution (Evans §2.2.1, Theorem 1)

A radial function $v(x) = F(r)$, $r = |x| > 0$, satisfies

$$\Delta v = F''(r) + \frac{n-1}{r}F'(r).$$

Thus harmonicity off zero gives $(r^{n-1}F')' = 0$. Integration yields a constant plus r^{2-n} if $n \geq 3$, or a constant plus $\log r$ if $n = 2$. Choose the constants so that the inward flux is one:

$$\Phi(x) = \begin{cases} [n(n-2)\omega_n]^{-1}|x|^{2-n}, & n \geq 3, \\ -(2\pi)^{-1}\log|x|, & n = 2. \end{cases} \qquad -\Delta\Phi = \delta_0.$$

符号由通量固定

我们求解 $-\Delta u = f$，故要求 $-\int_{\partial B_r}\partial_\nu\Phi = 1$。若改用 Δ 作基本算子，基本解及 Green 公式的符号也要一起改。

Expanded proof (the distributional singularity).

The displayed function is locally integrable and harmonic away from zero, and $\partial_r\Phi = -1/(n\omega_n r^{n-1})$. For $\phi \in C_c^\infty$, apply Green's identity on a large ball containing its support, with $\overline{B}_\varepsilon$ removed:

$$\int_{|x|>\varepsilon}\Phi(-\Delta\phi) = \int_{\partial B_\varepsilon}(\Phi\partial_r\phi - \phi\partial_r\Phi)\,dS.$$

The first term tends to zero: it is $O(\varepsilon)$ for $n \geq 3$, and $O(\varepsilon|\log\varepsilon|)$ for $n = 2$. The second is the normalized spherical mean of ϕ, tending to $\phi(0)$. Local integrability permits passage in the volume integral. Hence $-\Delta\Phi = \delta_0$ as a distribution. □

For $f \in C_c^2(\mathbb{R}^n)$ put $u = \Phi * f$. Write $u(x) = \int \Phi(y) f(x-y)\,dy$ and differentiate f, not the singular kernel. On each compact set of x, all these integrals have common bounded support in y and are dominated by an integrable multiple of $|\Phi|$. Thus $u \in C^2$ and

$$-\Delta u(x) = \int \Phi(y)\,(-\Delta_y f(x-y))\,dy = f(x)$$

by the distribution identity just proved. This avoids an illegal unqualified differentiation of $D^2\Phi$ at zero. For $n = 1$, $\Phi(x) = -|x|/2$ has the same normalization.

2.2.2 Mean-value formulas (Evans §2.2.2, Theorems 2–3)

Write $\omega_n = |B_1|$ and $\sigma_{n-1} = |\partial B_1| = n\omega_n$. For $\overline{B}_r(x) \subset U$ define the *normalized* means

$$S_u(x,r) = \frac{1}{n\omega_n r^{n-1}} \int_{\partial B_r(x)} u\,dS, \qquad A_u(x,r) = \frac{1}{\omega_n r^n} \int_{B_r(x)} u\,dy.$$

The sphere is a boundary of dimension $n-1$; the ball has dimension n. These are two different averaging measures.

平均不是积分本身

球面平均除以面积，球体平均除以体积。对常数 $u = 1$，两者都必须等于 1；漏掉 r^{n-1} 或 r^n 就会破坏这一检验。

Theorem 2.1 (Mean-value identities)
If $u \in C^2(U)$ and $\Delta u = 0$, then

$$u(x) = S_u(x,r) = A_u(x,r) \quad \text{whenever } \overline{B}_r(x) \subset U.$$

More generally, for any C^2 function,

$$\partial_r S_u(x,r) = \frac{1}{n\omega_n r^{n-1}} \int_{B_r(x)} \Delta u(y)\,dy. \tag{M}$$

Expanded proof (all radius and surface factors).

1. Remove the moving surface before differentiating. Set $y = x + r\theta$ on the unit sphere. Its surface Jacobian is r^{n-1}, so

$$S_u(x,r) = \frac{1}{n\omega_n} \int_{\partial B_1} u(x + r\theta)\,dS_\theta.$$

On a fixed compact range of radii, Du is continuous and bounded. Differentiation under this fixed finite integral is legitimate:

$$\partial_r S_u(x,r) = \frac{1}{n\omega_n} \int_{\partial B_1} Du(x + r\theta)\cdot\theta\,dS_\theta.$$

Transforming back, $\theta = \nu(y)$ is the outward unit normal, giving

$$\partial_r S_u(x,r) = \frac{1}{n\omega_n r^{n-1}} \int_{\partial B_r(x)} Du\cdot\nu\,dS.$$

The divergence theorem on the smooth ball changes the flux into $\int_{B_r(x)} \Delta u$. This proves (M).

2. Identify the constant. If $\Delta u = 0$, the derivative in (M) vanishes for every admissible r. Hence the spherical mean is independent of radius. Continuity gives

$$|S_u(x,r) - u(x)| \le \sup_{|y-x|\le r} |u(y) - u(x)| \longrightarrow 0.$$

The constant is therefore exactly $u(x)$, not an unspecified constant.

3. Recover the ball average by adding spherical shells. Polar integration yields

$$\int_{B_r(x)} u = \int_0^r \int_{\partial B_\rho(x)} u \, dS \, d\rho = \int_0^r n\omega_n \rho^{n-1} u(x) \, d\rho = \omega_n r^n u(x).$$

Divide by the volume. This also proves, for general continuous u,

$$A_u(x,r) = \frac{n}{r^n} \int_0^r \rho^{n-1} S_u(x,\rho) \, d\rho. \tag{SB}$$

□

图 T2.M：球体由球壳相加



shell volume
$n\omega_n \rho^{n-1} \, d\rho$

先对每个半径做球面平均，再乘球壳体积并积分。体积平均的径向权重不是均匀的。

Can one recover spheres from balls? Yes. Differentiating $r^n A_u(x,r)$ in (SB) gives

$$S_u(x,r) = A_u(x,r) + \frac{r}{n} \partial_r A_u(x,r).$$

Consequently a continuous function with the ball-mean identity for every small radius also has the sphere-mean identity. A single radius at a single center is not sufficient.

方程如何变成平均

关键中介是通量：$\Delta u = 0$ 令每个球的总法向通量为零，通量又恰好是球面平均对半径的导数。故平均不随尺度变化。

Theorem 2.2 (Converse, including a continuous version)
If $u \in C^2(U)$ has either mean identity for every sufficiently small ball at each center, then $\Delta u = 0$. If $u \in C(U)$ has this property at every center and all locally admissible small radii, it is smooth and harmonic.

Expanded proof.

For C^2 functions Taylor-expand at x with a uniform $o(r^2)$ error. Symmetry of the sphere cancels linear and mixed terms; its normalized second moments are $\mathbb{E}(\theta_i\theta_j) = \delta_{ij}/n$. The ball has moments $\mathbb{E}(z_i z_j) = r^2\delta_{ij}/(n+2)$. Thus

$$S_u(x,r) = u(x) + \frac{r^2}{2n}\Delta u(x) + o(r^2),$$

$$A_u(x,r) = u(x) + \frac{r^2}{2(n+2)}\Delta u(x) + o(r^2). \tag{T}$$

An exact identity for arbitrarily small r forces $\Delta u(x) = 0$.

For a merely continuous u, first turn ball means into sphere means using (SB), if necessary. Take a radial smooth mollifier of mass one, supported in B_ε. In polar coordinates,

$$(\rho_\varepsilon * u)(x) = \int_0^\varepsilon \rho_\varepsilon(r) \int_{\partial B_r(x)} u \, dS \, dr = u(x).$$

The convolution is smooth on the smaller domain; hence u is smooth there and throughout U. Now apply the C^2 converse. This proves the smoothness assertion without differentiating an initially nonsmooth function. □

Subharmonic and superharmonic functions. If $\Delta v \geq 0$, (M) implies $S_v(x,r) \geq v(x)$ and then $A_v(x,r) \geq v(x)$. If $\Delta v \leq 0$, reverse both inequalities. For $-\Delta u = f$, integrating (M) and changing the order of integration gives, for $n \geq 3$,

$$u(x) = S_u(x,r) + \int_{B_r(x)} \frac{|y-x|^{2-n} - r^{2-n}}{n(n-2)\omega_n} f(y)\, dy. \tag{P}$$

Indeed $\int_{|y-x|}^r \rho^{1-n}\, d\rho = (|y-x|^{2-n} - r^{2-n})/(n-2)$. For $n = 2$ replace the kernel by $(2\pi)^{-1}\log(r/|y-x|)$. The kernel is nonnegative: a positive source raises $u(x)$ above its boundary average under the convention $-\Delta u = f$.

Concrete checks. In two dimensions, $u(y) = y_1^2 - y_2^2$ and $u(y) = y_1 y_2$ have $\Delta u = 0$, so both means reproduce $u(x)$ at every center. For $v(y) = |y|^2$,

$$S_v(0,r) = r^2, \qquad A_v(0,r) = \frac{n}{n+2} r^2, \qquad v(0) = 0.$$

For $n = 1$, the sphere consists of two endpoints: $S_u(x,r) = [u(x-r) + u(x+r)]/2$. Harmonic functions are affine, and the same identities remain valid with $\omega_1 = 2$.

2.2.3 Consequences of averaging (Evans §2.2.3)

Theorem 2.3 (Strong maximum principle; source Theorems 4–5)
If U is bounded and $u \in C^2(U) \cap C(\overline{U})$ is harmonic, its maximum and minimum are attained on ∂U. If U is connected and a maximum is attained inside, u is constant. Consequently the classical Dirichlet problem $-\Delta u = f$, $u = g$ has at most one solution in this class.

二阶误差读出 Laplace

普通函数的平均一般不等于中心值；首个对称性允许的误差就是 $r^2\Delta u$。调和性让这一误差及全部更高阶平均误差精确抵消。

别混同一元中值定理

这里是积分平均精确重现中心值，不是声称“存在某个中间点”。后面的热球公式也保持中心值，但必须改变平均的几何区域和权重。

Expanded proof.

Let $M = \max_{\overline{U}} u$ and suppose $u(x) = M$ at an interior point. The ball identity gives $\int_{B_r(x)} (M - u) = 0$. The integrand is nonnegative and continuous. If it were positive at one point, continuity would give a positive integral on a smaller ball, a contradiction. Thus $u = M$ throughout $B_r(x)$. The set $\{u = M\}$ is open and relatively closed in U. On a connected component containing an interior maximum it is the whole component, and continuity transfers this value to that component's boundary, which lies in ∂U. This proves the boundary maximum statement even if U has several components. Apply the argument to $-u$ for minima. The difference of two Dirichlet solutions is harmonic with zero boundary values, so its maximum and minimum are both zero. □

正权平均的等号条件

平均值达到所有被平均数的上界，意味着它们几乎处处都达到该上界；连续性再把“几乎处处”提升为处处。热球的强最大值原理使用同一机制。

Smoothness and derivative estimates (source Theorems 6–7). Radial convolution equals u by the mean identity, as proved in §2.2.2, so harmonic functions are C^∞. Fix a radial mollifier ρ with support in B_1. For $B_r(x) \Subset U$, use a convolution radius $r/2$ near x:

$$|D^\alpha u(x)| = \left| \int D^\alpha \rho_{r/2}(x - y) u(y)\, dy \right| \le C_{\alpha,n} r^{-n-|\alpha|} \|u\|_{L^1(B_r(x))}. \tag{D}$$

All derivatives land on the smooth kernel; its scaling explains the power of r. With a first derivative the same argument gives $\|Du\|_{L^\infty(B_{R-\delta})} \le C_n \delta^{-1} \|u\|_{L^\infty(B_R)}$.

> **Theorem 2.4** (Liouville and bounded Poisson representation; source 8–9)
> *A bounded harmonic function on $\mathbb{R}^n$ is constant. For $n \ge 3$ and $f \in C_c^2$, every bounded solution of $-\Delta u = f$ is $\Phi * f + C$.*

Expanded proof.

Apply (D) for $|\alpha| = 1$ on $B_R(x)$: $|Du(x)| \le C_n R^{-1} \|u\|_\infty \to 0$ as $R \to \infty$. Thus $Du = 0$ and connectedness makes u constant. For $n \ge 3$, $\Phi * f$ is continuous and tends to zero at infinity, hence bounded. Subtract it from u and apply Liouville. For $n = 2$, the logarithmic potential need not be bounded at infinity, so that final step cannot be asserted with the same hypotheses. □

> **Theorem 2.5** (Analyticity; source Theorem 10)
> *Harmonic functions are real analytic in the interior.*

Expanded proof (quantitative iteration rather than smoothness alone).

Fix $B_R(x_0) \Subset U$, with $M = \|u\|_{L^\infty(B_R)}$. Every derivative of u is harmonic, because derivatives commute with Δ. Iterate the first-derivative

estimate k times on concentric balls whose successive radii differ by $R/(2k)$. For any $|\alpha| = k$ this gives

$$\|D^\alpha u\|_{L^\infty(B_{R/2})} \le M(2C_n k/R)^k \le M(2eC_n/R)^k k!.$$

The last inequality follows from $\log(k!) = \sum_{j=1}^{k} \log j \ge \int_1^k \log s\, ds \ge k\log k - k$. For h small enough that the segment from x_0 to $x_0 + h$ stays in $B_{R/2}$, apply one-dimensional Taylor's theorem to $q(t) = u(x_0 + th)$. Its kth derivative is

$$q^{(k)}(t) = \sum_{|\alpha|=k} \frac{k!}{\alpha!} h^\alpha D^\alpha u(x_0 + th).$$

Dividing the remainder by $k!$ and using $\sum_{|\alpha|=k} |h^\alpha|/\alpha! = |h|_1^k/k!$ bounds it by $M(2eC_n|h|_1/R)^k$, tending to zero for sufficiently small h. Thus the multivariable Taylor series equals u near x_0. □

Theorem 2.6 (Harnack; source Theorem 11)
For connected $V \Subset U$ and nonnegative harmonic u, $\sup_V u \le C_{U,V,n} \inf_V u$.

Expanded proof.

Choose $\rho > 0$ with every $B_{2\rho}(z)$, $z \in V$, inside U. If $x, y \in V$ and $|x - y| \le \rho$, then $B_\rho(y) \subset B_{2\rho}(x)$ and positivity gives

$$u(x) = \frac{1}{|B_{2\rho}|} \int_{B_{2\rho}(x)} u \ge 2^{-n} \frac{1}{|B_\rho|} \int_{B_\rho(y)} u = 2^{-n} u(y).$$

Exchange x, y for the other inequality. Compactness gives a finite $\rho/4$ net of V. Connectedness permits a chain between any two net points with successive centers separated by at most ρ; one may obtain this by covering a path in V and deleting loops in the finite cover graph. There is a uniform finite bound on the chain length. Multiplying the local factors proves the global comparison. If one value is zero, the same inequalities force every value in V to be zero, so the statement includes that case. □

光滑不等于解析

C^∞ 只说明每阶导数存在；解析还需控制高阶导数的增长。这里的 $k!$ 型估计让 Taylor 余项趋零，不能省略这一环。

2.2.4 Green and Poisson kernels (Evans §2.2.4)

For sufficiently smooth u, v, the two Green identities are

$$\int_U Du \cdot Dv = -\int_U v\Delta u + \int_{\partial U} v\partial_\nu u,$$

$$\int_U (u\Delta v - v\Delta u) = \int_{\partial U} (u\partial_\nu v - v\partial_\nu u).$$

The second is the difference of the first identity in the two orders.

Boundary correction. For fixed $x \in U$, suppose a harmonic corrector h_x with $h_x(y) = \Phi(y - x)$ on ∂U has been constructed. Set $G(x, y) = \Phi(y - x) - h_x(y)$. Then

$$-\Delta_y G(x, y) = \delta_x, \qquad G(x, y) = 0 \quad (y \in \partial U).$$

General-domain existence of this corrector is not proved by the notation; it belongs to Dirichlet existence theory later in the book. The following representation is conditional on such a sufficiently regular Green function. Explicit examples below avoid that existence issue.

Green 函数是修正后的点源

基本解有正确的点源奇性，却通常不满足边界条件。减去调和修正项保留奇性、消掉边界值；“定义修正项”本身不等于证明它存在。

Theorem 2.7 (Representation and symmetry; source Theorems 12–13)
For $-\Delta u = f$, $u = g$ on the boundary,

$$u(x) = \int_U G(x, y) f(y)\, dy - \int_{\partial U} g(y)\, \partial_{\nu_y} G(x, y)\, dS_y. \qquad \text{(G)}$$

Furthermore $G(x, y) = G(y, x)$.

Expanded proof.

Apply the second Green identity on $U \setminus \overline{B}_\varepsilon(x)$ to u and $G(x, \cdot)$. Away from x, $\Delta G = 0$. On the outer boundary $G = 0$, leaving $\int g \partial_\nu G$. On the inner boundary the singular normal derivative has total flux one, with normal directed into the removed ball; the resulting term tends to $u(x)$. Terms involving $G \partial_\nu u$ tend to zero at rates $O(\varepsilon)$ or $O(\varepsilon |\log \varepsilon|)$. Thus $\int_U Gf = \int_{\partial U} g \partial_\nu G + u(x)$, giving (G).

For symmetry, apply the same second identity to $G(x, \cdot), G(z, \cdot)$ with small disjoint balls about x, z removed. Both are harmonic on the remaining domain and zero on the outer boundary. At the sphere around x, the term with the singular normal derivative tends to $G(z, x)$ and the opposite product tends to zero; the sphere around z gives $-G(x, z)$. Their sum is zero, proving symmetry. □

The upper half-space (source Theorem 14). For $x_n > 0$ let $x^* = (x', -x_n)$. Reflection gives $G(x, y) = \Phi(y - x) - \Phi(y - x^*)$. The singularity of the image lies outside the domain, and the two distances agree on $y_n = 0$. Its boundary normal derivative gives

$$P(x, z) = \frac{2x_n}{n\omega_n (|x' - z|^2 + x_n^2)^{n/2}}, \quad z \in \mathbb{R}^{n-1}.$$

$$u(x) = \int_{\mathbb{R}^{n-1}} P(x, z) g(z)\, dz.$$

The kernel is harmonic in x, positive, and has mass one. To check the mass without imposing boundary decay, put $z = x' + x_n w$; the remaining integral is $[2/(n\omega_n)] \int (1 + |w|^2)^{-n/2}\, dw = 1$. Its value follows by

polar coordinates, or the Gaussian integral and Gamma substitution. For bounded continuous g, split the boundary integral into $|z - x_0'| < \delta$ and its complement. Uniform continuity near x_0' controls the first error; the latter kernel mass tends to zero as $x \to (x_0', 0)$. This proves boundary recovery and, by differentiation on compact interior sets, harmonicity. Uniqueness here is among bounded solutions: a bounded zero-boundary harmonic function extends oddly across the plane, is harmonic by the reflection argument in Problem 10, and is constant by Liouville, hence zero.

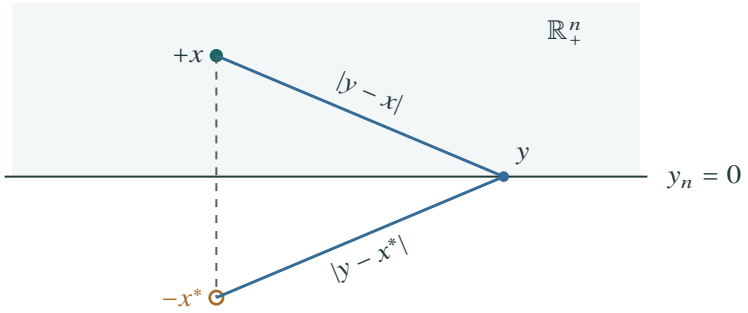


Diagram T2.2. A reflected point source enforces the boundary value. The solid source at x is inside the upper half-space; the image at x^* is outside and has the opposite sign. Every boundary point y is equidistant from the pair, so their potentials cancel there. The labels indicate source signs, not signed coordinate locations.

The ball (source Theorem 15). For $n \geq 3$, $0 < |x| < R$, put $x^* = R^2 x/|x|^2$. An explicit Green function is

$$G(x, y) = \Phi(y - x) - (R/|x|)^{n-2} \Phi(y - x^*).$$

On $|y| = R$, $|y - x^*| = (R/|x|)|y - x|$, so the boundary value is zero. At $x = 0$ the formula has the limit $\Phi(y) - \Phi(R)$. In two dimensions use instead

$$G(x, y) = \frac{1}{2\pi} \log \frac{|x|\,|y - x^*|}{R|y - x|},$$

with the corresponding center limit $(2\pi)^{-1} \log(R/|y|)$. Differentiating in the boundary normal gives, for every $n \geq 2$,

$$P_R(x, y) = \frac{R^2 - |x|^2}{n\omega_n R|x - y|^n}, \qquad u(x) = \int_{\partial B_R} P_R(x, y) g(y)\, dS_y. \tag{PB}$$

Expanded proof (why the boundary formula solves the problem).

The constructed Green function and (G) applied to $u = 1$ give $\int_{\partial B_R} P_R = 1$. This uses a known constant harmonic function, so does

not assume existence for arbitrary data. For $|y| = R$, direct differentiation yields

$$\Delta_x\big[(R^2 - |x|^2)|x-y|^{-n}\big] = 2n(R^2 - |y|^2)|x-y|^{-n-2} = 0.$$

Thus the integral is harmonic; all differentiations are dominated on compact subsets separated from the sphere. For $x \to y_0 \in \partial B_R$ split $g(y) - g(y_0)$ into a small spherical neighborhood and its complement. Positivity and mass one control the near part by the modulus of continuity. On the complement the denominator stays bounded below while $R^2 - |x|^2 \to 0$, so the far part tends to zero. This proves the boundary data. The maximum principle gives uniqueness. At $x = 0$, P_R is constant and (PB) reduces exactly to the spherical mean identity. □

2.2.5 Energy and Dirichlet's principle (Evans §2.2.5, Theorems 16–17)

Let U be bounded and smooth enough for the following integrations. For two $C^2(\overline{U})$ solutions of the same Dirichlet problem, their difference w has $\Delta w = 0$ and $w|_{\partial U} = 0$. Green's first identity gives $\int_U |Dw|^2 = -\int_U w\Delta w + \int_{\partial U} w\partial_\nu w = 0$. Thus w is constant on each component, and its boundary values make it zero.

For $-\Delta u = f$, $u = g$, define

$$J(v) = \int_U \left(\tfrac{1}{2}|Dv|^2 - fv\right) dx.$$

For $v = u + w$ with $w|_{\partial U} = 0$,

$$J(v) - J(u) = \int_U (Du\cdot Dw - fw) + \tfrac{1}{2}\int_U |Dw|^2 = \tfrac{1}{2}\int_U |Dw|^2 \geq 0.$$

Here the linear term vanishes by Green's identity and the PDE. Conversely, if a smooth u minimizes J for its boundary data, test $v = u + \varepsilon\phi$ with $\phi \in C_c^\infty(U)$ and both signs of ε. Differentiating the quadratic polynomial at zero gives $\int Du\cdot D\phi = \int f\phi$, hence $-\Delta u = f$. This proves the equivalence in the smooth class; existence of a minimizer in a Sobolev class uses the later functional-analytic theory.

中心平均与偏心平均

从球心看，边界各方向权重相同；从偏心点看，靠近该点的边界权重更大，变成 Poisson 核。球面均值是 Poisson 表示的球心特例。

2.3 Heat Equation

2.3.1 The Gaussian kernel and Duhamel (Evans §2.3.1)

Heat balance with unit diffusivity is $u_t = \Delta u$. The scaling $x \mapsto \lambda x$, $t \mapsto \lambda^2 t$ suggests a mass-preserving kernel $t^{-n/2}F(x/\sqrt{t})$. Substitution gives $\Delta F + \frac{1}{2}z\cdot DF + \frac{n}{2}F = 0$. Taking $F(z) = Ce^{-|z|^2/4}$ satisfies this equation;

从第二章走向第五章

能量只含一阶导数，所以弱形式 $\int Du\cdot D\phi = \int f\phi$ 自然进入 H^1。第二章的显式公式与第五章的函数空间方法在这里接上。

the product of one-dimensional Gaussian integrals fixes $C = (4\pi)^{-n/2}$. Thus

$$K(x,t) = (4\pi t)^{-n/2} e^{-|x|^2/(4t)} \quad (t>0), \qquad K = 0 \quad (t \le 0).$$

The value at the single point $(0,0)$ is interpreted distributionally. Direct differentiation verifies

$$K_t = \left(-\frac{n}{2t} + \frac{|x|^2}{4t^2}\right) K, \qquad \Delta K = \left(-\frac{n}{2t} + \frac{|x|^2}{4t^2}\right) K.$$

Also $K > 0$, $\int K(x,t)\,dx = 1$, and $\int_{|x|>\delta} K(x,t)\,dx \to 0$ as $t \downarrow 0$. The last assertion follows from $x = \sqrt{t}\,z$ and the integrable Gaussian tail.

Theorem 2.8 (Source Theorem 1)
For bounded continuous g,

$$u(x,t) = \int_{\mathbb{R}^n} K(x-y,t) g(y)\,dy$$

is smooth for $t > 0$, solves the heat equation, and tends to $g(x_0)$ as $(x,t) \to (x_0, 0)$.

Expanded proof.

On compact subsets of $t > 0$, every kernel derivative is a polynomial times a Gaussian with an integrable uniform majorant. Differentiate under the integral and use $K_t = \Delta K$. For initial convergence use mass one:

$$u(x,t) - g(x_0) = \int K(x-y,t)\,[g(y) - g(x_0)]\,dy.$$

Given $\varepsilon > 0$, continuity makes the bracket smaller than ε for $|y - x_0| < \delta$. For $|x - x_0| < \delta/2$, the far part is bounded by $2\|g\|_\infty$ times a Gaussian tail outside $B_{\delta/2}(x)$, which tends to zero. This proves the joint limit. Completing the square in a product of two Gaussians gives $K_t * K_s = K_{t+s}$, the semigroup law. □

For a time-dependent source with zero initial data, Duhamel gives

$$v(x,t) = \int_0^t (K_{t-s} * f(\cdot,s))(x)\,ds.$$

For example assume f is smooth on the time interval, with uniformly compact spatial support and bounded indicated derivatives. Change to $\tau = t - s$ if needed to keep differentiations away from the singular upper endpoint. Moving spatial derivatives to f gives

$$v_t = f(x,t) + \int_0^t K_{t-s} * \Delta f(\cdot,s)\,ds, \qquad \Delta v = \int_0^t K_{t-s} * \Delta f(\cdot,s)\,ds.$$

抛物缩放为何是平方

空间二阶导数缩放为 λ^2，时间一阶导数也必须缩放为 λ^2 才保持方程。热球的高度因此正比 r^2，不是 r。

瞬时传播不等于大量传播

$K(x,t) > 0$ 对任意 x、$t > 0$ 成立，所以非零非负初值立即在全空间产生正值；远处的影响可极小，但不是严格为零。

The endpoint term uses $K_\varepsilon * f(\cdot,t) \to f(\cdot,t)$. The bounds on $f, D^2 f$ provide domination, proving $Lv = f$ and $v(\cdot,0) = 0$. Adding $K_t * g$ solves the full initial-value problem (source Theorem 2). We use these explicit smoothness assumptions rather than claim that arbitrary continuous forcing gives classical second derivatives without further justification.

2.3.2 Heat-ball mean values (Evans §2.3.2, Theorem 3)

We now expand the connection with harmonic means. The region, the time orientation, and the weight are all essential.

1. The kernel determines the geometry

Let $K(z,\tau) = (4\pi\tau)^{-n/2} e^{-|z|^2/(4\tau)}$, $\tau > 0$. For a target (x,t) define

$$E_r(x,t) = \{(y,s) : s < t,\ K(x-y, t-s) > r^{-n}\}.$$

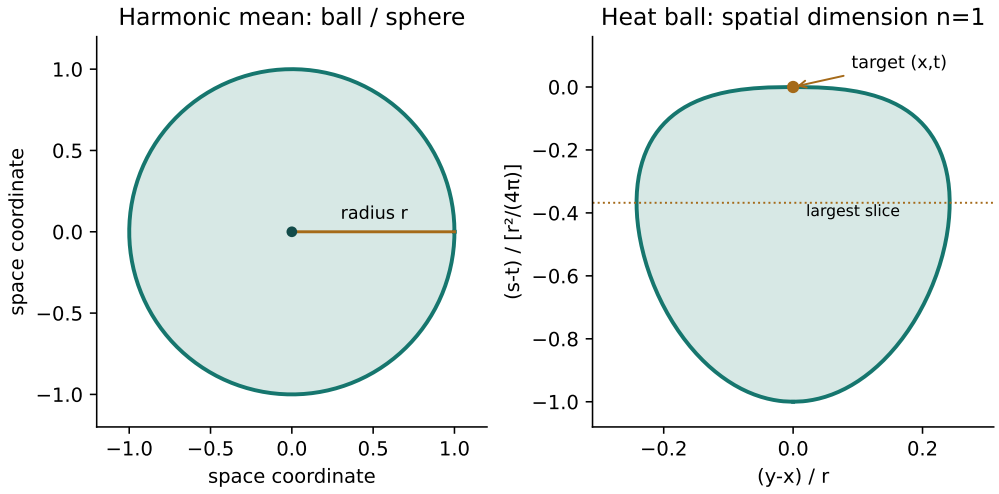
The open or closed choice on the ordinary boundary does not affect the integral. Put $\tau = t - s$ and $a = r^2/(4\pi)$. Taking logarithms gives

$$0 < \tau < a, \qquad |y - x| < R_r(\tau), \qquad R_r(\tau)^2 = 2n\tau \log(a/\tau). \quad \text{(HB)}$$

Each past slice is an ordinary spatial ball. Its radius vanishes at both tips and is largest at $\tau = a/e$:

$$\max R_r(\tau) = r\sqrt{\frac{n}{2\pi e}}.$$

Thus r is a kernel-level scale, not the largest spatial radius.



Editorial figure. Left: an elliptic ball and its boundary. Right: a one-dimensional spatial section of a heat ball, with time increasing toward the target. Spatial and temporal axes use different scales.

热球为何朝向过去

核 $K(x-y, t-s)$ 只在 $s < t$ 有意义。热球下端在 $t - r^2/(4\pi)$，上端是目标时刻 t；它不是以 (x,t) 为对称中心的时空球。

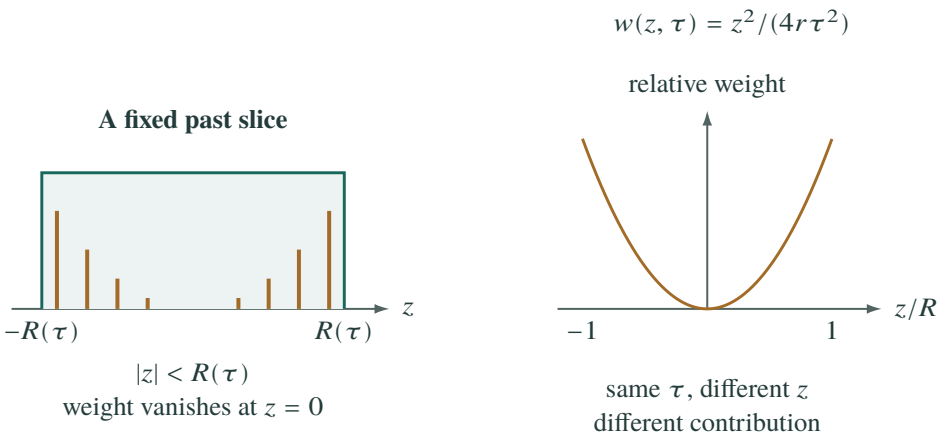


Diagram T2.3. A heat ball carries a nonuniform averaging measure. In one spatial dimension, fix a past time $\tau = t - s$. The blue band marks the allowed interval, while orange heights illustrate the factor $z^2/(4r\tau^2)$. The right panel shows its quadratic spatial dependence up to a fixed positive factor. This is a slice of the full space–time probability measure, not a separately normalized conditional density.

2. Statement and full proof

Theorem 2.9 (Watson–Evans mean-value formula)
If $u \in C^{2,1}$ solves $Lu = u_t - \Delta u = 0$ on a neighborhood of $\overline{E}_r(x,t)$, then

$$\boxed{u(x,t) = \frac{1}{4r^n} \iint_{E_r(x,t)} u(y,s)\, \frac{|x-y|^2}{(t-s)^2}\, dy\, ds.} \tag{H}$$

The nonnegative measure

$$d\mu_r(y,s) = \frac{\mathbf{1}_{E_r(x,t)}}{4r^n}\, \frac{|x-y|^2}{(t-s)^2}\, dy\, ds$$

has mass one, so this is $u(x,t) = \mathbb{E}_{\mu_r}[u(Y,S)]$.

Expanded proof (including the source's omitted normalization).

A. Compute the mass exactly. At fixed τ, $\int_{B_R} |z|^2\, dz = n\omega_n R^{n+2}/(n+2)$. With $b = n/2$ and $a = r^2/(4\pi)$, polar integration gives

$$\begin{aligned} I_r &:= \iint_{E_r(0,0)} \frac{|y|^2}{s^2}\, dy\, ds \\ &= \frac{n\omega_n}{n+2}(2n)^{b+1} \int_0^a \tau^{b-1} [\log(a/\tau)]^{b+1}\, d\tau. \end{aligned}$$

Set $q = \log(a/\tau)$, so $\tau = ae^{-q}$. Then

$$\int_0^a \tau^{b-1} [\log(a/\tau)]^{b+1}\, d\tau = a^b \int_0^\infty e^{-bq} q^{b+1}\, dq = a^b \frac{\Gamma(b+2)}{b^{b+2}}.$$

区域和权重缺一不可

均匀热球抽样使用 $dy\, ds/|E_r|$；这里使用 $|x-y|^2/[4r^n(t-s)^2]$。即使区域画对了，丢掉权重仍然不是均值定理。

Here $\Gamma(c) = \int_0^\infty z^{c-1}e^{-z}\,dz$; the last equality uses $z = bq$. Substitute $\omega_n = \pi^b/\Gamma(b+1)$, $a^b = r^n/(4\pi)^b$, and $\Gamma(b+2) = (b+1)\Gamma(b+1)$. Cancellation yields

$$I_r = 4r^n, \qquad \mu_r(E_r) = 1.$$

This also proves integrability of the seemingly singular top-tip weight for every $n \geq 1$.

B. Differentiate after fixing the domain. Translate the target to $(0,0)$; thus integration times s are negative. Scaling of K gives $E_r = \{(ry, r^2 s) : (y,s) \in E_1\}$. Define

$$\phi(r) = r^{-n} \iint_{E_r} u(y,s)\frac{|y|^2}{s^2}\,dy\,ds = \iint_{E_1} u(ry, r^2 s)\frac{|y|^2}{s^2}\,dy\,ds.$$

The Jacobian is r^{n+2} and the weight contributes r^{-2}, explaining the cancellation. First suppose u smooth. Differentiate on E_1, then scale back:

$$\phi'(r) = A + B,$$

$$A = r^{-n-1}\iint_{E_r}(y\cdot Du)\frac{|y|^2}{s^2}, \qquad B = r^{-n-1}\iint_{E_r} 2u_s\frac{|y|^2}{s}. \tag{AB}$$

Bounded derivatives on compact sets and the integrable weights justify this operation for radii in a compact admissible interval.

C. Find a boundary-vanishing function. Set

$$\psi_r = \log\left(r^n K(y,-s)\right) = n\log r - \frac{n}{2}\log(-4\pi s) + \frac{|y|^2}{4s}.$$

It is positive inside E_r, zero on its lateral boundary, and

$$D_y\psi_r = \frac{y}{2s}, \qquad (\psi_r)_s = -\frac{n}{2s} - \frac{|y|^2}{4s^2}. \tag{L}$$

Since $2|y|^2/s = 4y\cdot D\psi_r$, spatial integration by parts gives

$$B = -4r^{-n-1}\iint_{E_r}\psi_r(nu_s + y\cdot Du_s).$$

Integrate the last term in time and insert (L):

$$\begin{aligned} B &= r^{-n-1}\iint_{E_r}[-4n\psi_r u_s + 4(\psi_r)_s\, y\cdot Du] \\ &= -4nr^{-n-1}\iint_{E_r}\psi_r u_s - 2nr^{-n-1}\iint_{E_r}\frac{y\cdot Du}{s} - A. \end{aligned}$$

The A terms cancel. Use $y/s = 2D\psi_r$ and integrate spatially once more:

$$\boxed{\phi'(r) = 4nr^{-n-1}\iint_{E_r}\psi_r(\Delta u - u_s).} \tag{R}$$

For a caloric function the right side is zero, so ϕ is constant.

D. Check both tips and the regularity assumption. Zero lateral values alone do not justify ignoring the singular tip. First truncate to $\varepsilon < \tau < a - \eta$. The lateral terms vanish because $\psi_r = 0$. At the top cap, $R = O(\sqrt{\tau \log(a/\tau)})$ and

$$\int_{B_R} |y| \psi_r \, dy \le C R^{n+1} \log(a/\tau) \longrightarrow 0.$$

This controls the time-boundary term containing $y \cdot Du$. At the lower cap, $R \to 0$ and $\psi_r \to 0$, so that term also vanishes. The weight remaining near the upper tip is bounded by a constant times

$$\int_0^{\varepsilon} \tau^{n/2-1} [\log(a/\tau)]^{n/2+1} \, d\tau \longrightarrow 0.$$

The differentiated terms in (AB) and those with ψ_r have the same integrable powers or better ones. Let $\varepsilon, \eta \downarrow 0$.

For only $C^{2,1}$ regularity, convolve in space-time on a slightly larger neighborhood. The constant-coefficient operator L commutes with convolution, so smooth approximations still solve the equation where needed. They converge uniformly on the compact region; the integrable weight passes the established mean identity to u. This avoids requiring mixed third derivatives in the original function. At a terminal cylinder time, take limits from interior target times.

E. Identify the constant by shrinking. The heat ball lies in $|y| \le C_n r$, $|s| \le r^2/(4\pi)$. Uniform continuity and the mass computation imply $\phi(r) \to 4u(0,0)$ as $r \downarrow 0$. Therefore $\phi(r) = 4u(0,0)$ for every admissible radius. Divide by four and restore the target (x,t). □

3. The precise connection to elliptic averages

For general smooth u let $M_r[u] = \phi(r)/4$. The preceding computation, before imposing the PDE, gives

$$\frac{d}{dr} M_r[u](x,t) = -\frac{n}{r^{n+1}} \iint_{E_r(x,t)} \log\left(r^n K(x-y, t-s)\right) Lu(y,s) \, dy \, ds. \tag{HR}$$

Thus $Lu \le 0$ implies $M_r[u]$ nondecreasing, and shrinking r gives $u(x,t) \le M_r[u](x,t)$. Reverse the signs for supersolutions. This parallels $\Delta v \ge 0$ and $\partial_r S_v \ge 0$ in (M).

对数核为何出现

ψ_r 的零边界值消掉侧边界项；空间梯度产生平均权重；时间导数让缩放导数的两项抵消。这三种作用共同决定热球的定义。

A counterexample to unweighted replacements. The polynomial $u(y,s) = |y-x|^2 + 2n(s-t)$ has $Lu = 0$ and $u(x,t) = 0$. At time t its uniform spatial ball average is $nr^2/(n+2) > 0$. On $B_r(x) \times (t-r^2, t)$ its uniform space-time average is $nr^2/(n+2) - nr^2 \neq 0$. A symmetric Euclidean space-time ball cancels the time term but still leaves a positive spatial square. None of these is the required mean.

Stationary solutions. If $u(y,s) = h(y)$, then $Lu = 0$ is exactly $\Delta h = 0$. At fixed τ, the heat-ball weight is radial in $y - x$. Integrate over spherical shells first: every spherical mean of h equals $h(x)$, and the remaining scalar integral is the total mass one. Thus heat-ball averaging of a stationary solution is an aggregate of elliptic sphere averages. A time-dependent caloric solution needs the past slices as well as the weight.

Moments and a converse: editorial supplement. There is an exact probability description of μ_r. Put $b = n/2$, $a = r^2/(4\pi)$. Let Q have Gamma density proportional to $q^{b+1}e^{-bq}$, $q > 0$, and set $\tau = ae^{-Q}$. Conditional on τ, choose a uniform direction and radius $\varrho = R_r(\tau)U^{1/(n+2)}$, with U uniform on $(0,1)$. Step A gives the time density after the logarithmic substitution. The conditional spatial radial density is proportional to ϱ^{n+1}, because the volume Jacobian contributes ϱ^{n-1} and the weight contributes ϱ^2. Hence this construction samples the actual weighted measure.

Gamma integration and symmetry yield

$$\mathbb{E}(Y_i - x_i) = 0, \qquad \mathbb{E}[(Y_i - x_i)(Y_j - x_j)] = 2\delta_{ij}\mathbb{E}\tau,$$

$$\mathbb{E}\tau = c_n r^2, \qquad c_n = \frac{1}{4\pi}\left(\frac{n}{n+2}\right)^{n/2+2} > 0.$$

For the second moment, conditional radial integration gives $\mathbb{E}(|Y-x|^2 \mid \tau) = [(n+2)/(n+4)]\,2n\tau Q$. Integrating the Gamma density then gives $\mathbb{E}|Y-x|^2 = 2n\mathbb{E}\tau$. Taylor expansion for $C^{2,1}$ functions therefore implies

$$M_r[u](x,t) = u(x,t) + c_n r^2(\Delta u - u_t)(x,t) + o(r^2). \qquad \text{(HT)}$$

The remainder is uniform on the shrinking region, and $\mathbb{E}(|Y-x|^2 + \tau) = O(r^2)$ controls its integral. An exact identity for all small radii at every center forces $Lu = 0$. Compare this with the elliptic expansion (T).

两个次解的符号

椭圆次调和是 $\Delta v \geq 0$，即 $-\Delta v \leq 0$；抛物次解是 $v_t - \Delta v \leq 0$。统一算子符号后，两者都是“中心不超过正权平均”。

最简单的公式自检

$|y-x|^2 + 2n(s-t)$ 的空间曲率与时间变化恰好平衡。正确权重必须把二者抵消；普通空间平均看不到时间补偿。

几何平均在检测算子

球面平均的二阶误差是 $r^2\Delta u/(2n)$；热球平均的二阶误差是 $c_n r^2(\Delta u - u_t)$。两种几何平均分别检测各自的 PDE。

Do not confuse this with Gaussian averaging. At a fixed earlier time s, the whole-space solution can be written $u(x,t) = \int K(x-y,t-s)u(y,s)\,dy$ in a class where the Cauchy problem is unique. That is a global spatial Gaussian average on one time slice. Heat-ball averaging is local in space-time and uses many past slices. The fundamental solution is the same; the two averaging measures differ.

2.3.3 Maximum principles and regularity (Evans §2.3.3)

Write $U_T = U \times (0,T]$ and $\Gamma_T = (\overline{U} \times \{0\}) \cup (\partial U \times [0,T])$. The top $U \times \{T\}$ is not part of this parabolic boundary.

Theorem 2.10 (Source Theorems 4–5)
On bounded U, a continuous classical solution satisfies $\max_{\overline{U}_T} u = \max_{\Gamma_T} u$. If U is connected and the global maximum occurs at (x_0,t_0) with $x_0 \in U$, $t_0 > 0$, then u is constant on $\overline{U} \times [0,t_0]$. The initial-boundary problem has at most one classical solution for given forcing and parabolic boundary data.

Expanded proof.

For the weak maximum assertion set $v = u - \varepsilon t$. At a maximum in U_T, including the terminal time, the left time derivative is nonnegative and the spatial Laplacian is nonpositive. Thus $Lv \geq 0$, contradicting $Lv = -\varepsilon$. A maximum must lie on Γ_T. Let $\varepsilon \downarrow 0$.

For the strong assertion put $M = \max u$. The heat-ball mean at an interior maximum averages $u \leq M$ against a probability density strictly positive except on a negligible axis. Equality forces $u = M$ almost everywhere in the heat ball; continuity makes it equal everywhere. This propagates equality backward locally.

To reach a distant past point, use a polygonal spatial path in U and assign strictly decreasing times to its finitely many vertices. A small backward portion of each space-time line lies in a heat ball about its later endpoint: the spatial displacement is $O(\tau)$, whereas the heat-ball radius is of order $\sqrt{\tau \log(1/\tau)}$, which is larger for small τ. On each segment the earliest time of an already-constant terminal portion is attained by continuity. If it were not the segment's earliest endpoint, a small heat ball would extend it, a contradiction. Propagate through the finite chain. Every earlier interior point therefore has value M; continuity reaches time zero and the boundary. Finally apply the weak principle to both signs of the difference of two solutions for uniqueness. □

强最大值结论只向过去

内部时刻 t_0 达到全局最大值，会强迫此前恒定；不自动强迫此后恒定。边界条件在未来改变时，解也可以随后改变。

Whole-space uniqueness needs a growth class (source 6–7). Suppose $Lu \le 0$, $u(\cdot,0) \le 0$ and $u(x,t) \le Ae^{a|x|^2}$ for $0 \le t \le T$. Choose $b > a$ and the positive caloric barrier

$$V(x,t) = (1-4bt)^{-n/2} \exp\left(\frac{b|x|^2}{1-4bt}\right) \quad (0 \le t < 1/(4b)).$$

Its PDE follows from direct Gaussian-type differentiation. For fixed $\varepsilon > 0$, $u - \varepsilon V < 0$ on the lateral boundary of a sufficiently large ball, uniformly on a shorter interval $[0,\delta]$, $\delta < 1/(4b)$; the exponent $b > a$ ensures domination. Its initial values are nonpositive. The bounded-cylinder principle gives $u \le \varepsilon V$. Let the radius tend to infinity, then $\varepsilon \downarrow 0$. Repeat on consecutive time intervals of length δ, using the same growth bound, to cover $[0,T]$. This proves the whole-space maximum principle and, for $|u - \tilde{u}| \le Ae^{a|x|^2}$, uniqueness. Without a growth restriction the source discusses nonphysical zero-initial-data solutions; this draft does not use their existence as a substitute for specifying a uniqueness class.

Interior smoothing and derivative estimates (source 8–9). For a backward cylinder $Q_r = B_r(x_0) \times (t_0 - r^2, t_0]$ inside the domain, choose a smooth cutoff ζ equal to one on $Q_{3r/4}$, vanishing near the bottom and lateral boundary of Q_r. Then

$$L(\zeta u) = (\zeta_s - \Delta\zeta)u - 2D\zeta \cdot Du.$$

Duhamel and spatial integration by parts in the last term yield, on $Q_{r/2}$,

$$u(x,t) = \iint_{Q_r} \big[K(x-y,t-s)(\zeta_s + \Delta\zeta)(y,s) + 2D_y K(x-y,t-s) \cdot D\zeta(y,s)\big] u(y,s)\, dy\, ds. \tag{C}$$

Set the kernel to zero for $s \ge t$. Where cutoff derivatives do not vanish, either $t - s$ stays separated from zero or $|x - y|$ stays separated from zero. In the latter case every derivative of the Gaussian tends to zero faster than any power as $s \uparrow t$. Thus the combined kernel in (C) is smooth with bounded derivatives on the relevant compact sets, despite the original fundamental solution's singularity. A preliminary local convolution proves (C) for merely $C^{2,1}$ functions, then passes to the limit. Differentiating (C) now gives $u \in C^\infty$ locally.

On a unit cylinder, its kernel derivatives are bounded by constants $C_{\alpha,l}$. Rescale $v(z,\tau) = u(x_0 + rz, t_0 + r^2\tau)$ to obtain

$$\sup_{Q_{r/2}} |D_x^\alpha \partial_t^l u| \le C_{\alpha,l,n} r^{-n-2-|\alpha|-2l} \|u\|_{L^1(Q_r)}.$$

One time derivative costs two powers of spatial scale. For completeness, spatial analyticity can also be read from (C), using the elementary complex-analytic parameter-integral theorem and Cauchy estimates as additional prerequisites. Fix a time and replace the spatial variable by $z = x + iv$ in the Gaussian, using the polynomial $\sum_j (z_j - y_j)^2$ in its exponent. Where $t - s$ is small and cutoff derivatives are nonzero, $|x - y|$ is bounded below by some $d > 0$. For $|v| < d/2$ the real part of that polynomial remains positive, bounded below by a fixed multiple of d^2. The resulting Gaussian and its spatial derivatives have an integrable bound uniform on a small complex neighborhood. Where $t - s$ is bounded below there is no singularity at all. Thus (C) extends holomorphically in z near every interior spatial point. Cauchy's formula supplies factorial derivative bounds, proving real analyticity in space. This argument says nothing comparable about analyticity in time; smoothness alone does not establish that.

2.3.4 Energy and backward uniqueness (Evans §2.3.4, source 10–11)

Let w be the difference of two sufficiently smooth solutions with the same lateral boundary values. Then $w = 0$ on ∂U and

$$\frac{1}{2}\frac{d}{dt}\int_U w^2 = \int_U w\Delta w = -\int_U |Dw|^2.$$

Zero initial difference gives zero energy at every later time. This proves forward uniqueness and displays dissipation.

Backward uniqueness requires more than this decreasing-energy identity. Assume enough boundary and mixed derivative regularity for the next identities (smooth solutions up to the closed cylinder suffice). Set $e(t) = \int_U w^2$. Since $w_t = 0$ on the fixed lateral boundary,

$$e' = -2\int |Dw|^2, \qquad e'' = -4\int Dw \cdot Dw_t = 4\int (\Delta w)^2.$$

Cauchy–Schwarz in $\int |Dw|^2 = -\int w\Delta w$ gives $(e')^2 \le ee''$. Wherever $e > 0$, this means $(\log e)'' \ge 0$. If e were positive at t_1 and first vanished at $t_2 \le T$, convexity on $[t_1, t_2)$ would give, for $0 < \theta < 1$ and $b < t_2$,

$$e((1-\theta)t_1 + \theta b) \le e(t_1)^{1-\theta} e(b)^{\theta}.$$

Let $b \uparrow t_2$. The right side vanishes, contradicting positivity at the earlier interior time. Thus $e(T) = 0$ forces $e \equiv 0$.

切断函数移走奇点

把导数转移给核后，核奇点附近的切断导数全为零；远离空间中心时高斯又快速衰减。因此才可任意多次求导，而不是直接对奇异积分硬求导。

2.4 Wave Equation

2.4.1 Spherical means and explicit solutions (Evans §2.4.1)

The wave equation $u_{tt} - \Delta u = 0$ needs two initial data: $u(\cdot,0) = g$, $u_t(\cdot,0) = h$. Unlike the heat equation, it scales with $t \mapsto \lambda t$ when $x \mapsto \lambda x$.

One dimension: d'Alembert (source Theorem 1). Write $\xi = x + t$, $\eta = x - t$. Then $\partial_t^2 - \partial_x^2 = -4\partial_\xi \partial_\eta$. Thus $u_{\xi\eta} = 0$, and integration gives $u = F(\xi) + G(\eta)$. At $t = 0$, $F + G = g$ and $F' - G' = h$, whereas $F' + G' = g'$. Solving for F', G' and integrating yields

$$u(x,t) = \frac{g(x+t) + g(x-t)}{2} + \frac{1}{2}\int_{x-t}^{x+t} h(y)\,dy. \tag{DA}$$

For $g \in C^2$, $h \in C^1$, direct differentiation verifies the equation, $u(\cdot,0) = g$, and $u_t(\cdot,0) = h$. Only initial points in $[x-t, x+t]$ matter.

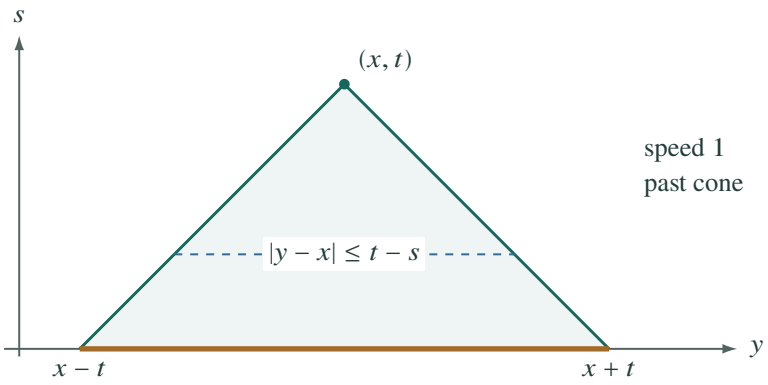


Diagram T2.4. The wave equation has a finite domain of dependence. For speed one in one dimension, the solution at (x,t) uses initial data only on $[x-t, x+t]$. At an intermediate time s, the relevant interval is $|y - x| \le t - s$. In higher dimensions this interval becomes a ball. A forced equation also accumulates sources from the past cone.

The common spherical-mean identity. Fix a spatial center x and set $M(r,t) = |\partial B_r|^{-1} \int_{\partial B_r(x)} u(y,t)\,dS$. The radius-flux computation used for harmonic means now gives

$$M_r = \frac{1}{n\omega_n r^{n-1}} \int_{B_r(x)} \Delta u, \qquad (r^{n-1} M_r)_r = r^{n-1} M_{tt}.$$

Thus M satisfies the Euler–Poisson–Darboux equation

$$M_{tt} = M_{rr} + \frac{n-1}{r} M_r, \qquad M(0,t) = u(x,t), \quad M_r(0,t) = 0. \tag{EPD}$$

逆向唯一不等于逆向稳定

知道最终状态可唯一确定过去，不代表微小最终误差只造成微小过去误差。热方程高频逆向增长仍会导致严重不稳定，第一章的例子依然适用。

两个初值各管什么

初始位移 g 生成两列反向平移的形状；初始速度 h 在区间内累积。热方程只需一个初值，而波动方程的二阶时间导数要求两个。

The second radius derivative differentiates the ball integral by shells; the wave equation changes Δu into u_{tt}.

Three dimensions: Kirchhoff. For $n = 3$, $W(r,t) = rM(r,t)$ satisfies $W_{tt} = W_{rr}$. Extend M evenly in r, hence W oddly. Its initial data are $rS_g(x,r)$ and $rS_h(x,r)$. Use (DA) for W and differentiate in r at zero. Since $W_r(0,t) = u(x,t)$, this gives

$$u(x,t) = \partial_t\big[tS_g(x,t)\big] + tS_h(x,t). \tag{KIR}$$

Equivalently

$$u(x,t) = \frac{1}{4\pi}\int_{\partial B_1}\left[g(x+t\theta) + tDg(x+t\theta)\cdot\theta + th(x+t\theta)\right]dS_\theta.$$

The latter fixed-sphere expression makes initial limits transparent. The formula solves the wave equation: by (EPD) applied to spatial spherical means of g or h, tS_h satisfies the wave equation in the variables (x,t); time differentiation handles the g term. At zero, $tS_h = th(x) + O(t^3)$ and $\partial_t(tS_g) = g(x) + O(t^2)$. These observations prove existence as well as the derivation from a hypothetical solution, for smooth data.

Two dimensions: descent from three dimensions. Make g, h independent of a third coordinate. Project the upper and lower hemispheres onto the disk. Each contributes surface Jacobian $t/\sqrt{t^2 - |y-x|^2}$. Formula (KIR) becomes

$$u(x,t) = \partial_t\left[\frac{1}{2\pi}\int_{B_t(x)}\frac{g(y)}{\sqrt{t^2-|y-x|^2}}\,dy\right] + \frac{1}{2\pi}\int_{B_t(x)}\frac{h(y)}{\sqrt{t^2-|y-x|^2}}\,dy. \tag{W2}$$

Use $y = x + tz$ before differentiating: the singularity $(1-|z|^2)^{-1/2}$ is integrable on the fixed unit disk. This justifies differentiation for sufficiently smooth data. Descent preserves the PDE because the third-coordinate derivative is zero; the three-dimensional initial limits give the desired two data.

All odd dimensions (source Theorem 2). For $n = 2k+1 \geq 3$, put $D_r = r^{-1}\partial_r$ and $\gamma_n = 1\cdot 3\cdots(n-2)$. Define

$$S_n(t)f = \frac{1}{\gamma_n}D_t^{k-1}\big[t^{n-2}S_f(x,t)\big].$$

Then

$$u(x,t) = \partial_t S_n(t)g + S_n(t)h. \tag{WO}$$

Here $S_n(t)$ denotes a solution operator, whereas $S_f(x,t)$ is the normalized spherical average defined earlier.

同一个通量公式三次出现

Laplace 方程令球面平均对半径不变；波动方程把同一半径变化联系到二阶时间导数；热方程则需要带方向的时空平均。不要因都出现球面而混为一谈。

降维为何出现权重

投影把球面压到圆盘，靠近圆周时球面近乎竖直，面积 Jacobian 变大，产生 $(t^2-|y-x|^2)^{-1/2}$。它是几何投影权重，不是热球的权重。

Expanded proof (the radial intertwining step).

Let $E = r\partial_r$ and $A_k F = D_r^{k-1}(r^{2k-1}F) = rP(E)F$, where $P(E) = \prod_{j=1}^{k-1}(E+2j+1)$; for $k=1$ the product is one. This factorization follows successively from $D_r(r^a F) = r^{a-2}(E+a)F$. The radial Laplacian is $r^{-2}E(E+2k-1)$. The polynomial identity

$$(E+1)P(E) = (E+2k-1)P(E-2)$$

therefore proves

$$(A_k F)'' = A_k\left(F'' + \frac{2k}{r}F'\right).$$

For $F(r) = S_f(x,r)$ the bracket equals $\Delta_x F$ by the divergence identity behind (EPD). Hence $A_k S_f$, now with $r = t$, solves the wave equation. It commutes with Δ_x since the operator acts only on r.

Spherical symmetry gives $S_f(x,t) = f(x) + O(t^2)$ with an even smooth extension at zero. The coefficient of $tf(x)$ in $A_k S_f$ is $P(0) = 3\cdot 5\cdots(2k-1) = \gamma_n$. All other terms start at t^3. Thus $S_n(t)f = tf + O(t^3)$, giving initial value zero and initial velocity f. Its time derivative has initial value f and initial velocity zero. This proves (WO) for smooth data and explains its constants. Finite differentiability follows by differentiating the fixed-sphere integrals; the source sufficient assumptions are $g \in C^{m+1}$, $h \in C^m$, $m = (n+1)/2$. □

All even dimensions (source Theorem 3). For even $n \geq 2$, put $\gamma_n = 2\cdot 4\cdots n$ and

$$B_f(x,t) = \frac{t^n}{\omega_n t^n}\int_{B_t(x)} \frac{f(y)}{\sqrt{t^2-|y-x|^2}}\,dy = \frac{1}{\omega_n}\int_{B_t(x)} \frac{f(y)}{\sqrt{t^2-|y-x|^2}}\,dy.$$

Set $S_n(t)f = \gamma_n^{-1} D_t^{(n-2)/2} B_f(x,t)$. The same formula $u = \partial_t S_n(t)g + S_n(t)h$ holds. To derive it, extend the data constantly in one extra coordinate, apply (WO) in dimension $n+1$, and project both hemispheres. Their Jacobian gives $2t/\sqrt{t^2-|y-x|^2}$; the sphere and ball area constants simplify to the displayed γ_n. The inherited PDE and initial limits prove it. For smooth data there is no endpoint issue; for the source finite regularity statement take $m = (n+2)/2$, $g \in C^{m+1}$, $h \in C^m$. When $n = 2$, this reduces to (W2).

For odd $n \geq 3$, only data on the sphere at distance t enter the formula, the sharp Huygens phenomenon. For even n, the whole ball enters and a trailing influence remains. Dimension one is an exception to the odd-dimensional sharp-front statement: (DA) includes an interval integral of h.

2.4.2 Forcing (Evans §2.4.2, Theorem 4)

The operator $S_n(t)$ above solves the homogeneous equation with initial displacement zero and velocity f. For smooth forcing set

$$v(t) = \int_0^t S_n(t-s)f(s)\,ds.$$

Since $S_n(0) = 0$ and $\partial_t S_n(0) = I$, differentiate twice:

$$v_t = \int_0^t \partial_t S_n(t-s)f(s)\,ds, \qquad v_{tt} = f(t) + \int_0^t \partial_t^2 S_n(t-s)f(s)\,ds.$$

The integrals in v_{tt} and Δv cancel. Thus $v_{tt} - \Delta v = f$, with both initial data zero. Add the homogeneous solution for general initial data.

2.4.3 Energy and finite propagation (Evans §2.4.3, Theorems 5–6)

For a smooth solution with fixed zero Dirichlet boundary values,

$$\mathscr{E}(t) = \frac{1}{2}\int_U (u_t^2 + |Du|^2), \qquad \mathscr{E}'(t) = \int_{\partial U} u_t \partial_\nu u = 0.$$

The boundary term vanishes because $u = 0$ there for all times implies $u_t = 0$ there. Apply this to the difference for uniqueness; zero initial energy gives $Du = u_t = 0$, and zero initial displacement removes constants.

For local propagation, consider the shrinking ball $B_{R-t}(x_0)$ and the cone energy

$$e(t) = \frac{1}{2}\int_{B_{R-t}(x_0)} (u_t^2 + |Du|^2).$$

Differentiating the moving domain contributes a negative surface term. The PDE and spatial integration by parts yield

$$e'(t) = \int_{\partial B_{R-t}} \left[u_t \partial_\nu u - \tfrac{1}{2}(u_t^2 + |Du|^2)\right] dS \le 0.$$

The inequality follows from $2u_t\partial_\nu u \le u_t^2 + |\partial_\nu u|^2 \le u_t^2 + |Du|^2$. If both initial data vanish on $B_R(x_0)$, $e(0) = 0$ and hence $e(t) = 0$ for $t < R$. Along each vertical segment inside the cone $u_t = 0$ and its initial value is zero, so $u = 0$ throughout the cone. Continuity gives the apex value when applicable.

2.5 Problems

Source: Evans §2.5, Problems 1–24, printed pp. 81–86 (PDF pp. 98–103). Statements are concisely restated, and solutions are editorial. Smoothness is assumed wherever a displayed differentiation requires it. Problems 3–5 and 17 are especially useful checks on the mean-value arguments.

2.5.1 Problems 1–8: transport and elliptic averages

1. Transport with a zeroth-order term. Solve $u_t + b \cdot Du + cu = 0$, $u(\cdot, 0) = g$.

Solution. Along $X(s) = x + (s-t)b$, the quantity $e^{cs}u(X(s), s)$ has derivative zero. Evaluate at $s = 0, t$ to obtain $u(x, t) = e^{-ct}g(x - tb)$. □

2. Rotation invariance of the Laplacian.

Solution. For an orthogonal matrix O, the Hessian of $v(x) = u(Ox)$ is $D^2v = O^T(D^2u)(Ox)O$. Its trace equals $\operatorname{tr}((D^2u)(Ox)OO^T) = \Delta u(Ox)$. Thus a harmonic u gives a harmonic v. □

3. The center value for Poisson's equation. Derive the center formula for $-\Delta u = f$ on B_r, $u = g$ on ∂B_r.

Solution. Integrate (M) from zero to r:

$$S_u(0, r) - u(0) = -\frac{1}{n\omega_n} \int_0^r \rho^{1-n} \int_{B_\rho} f(y)\, dy\, d\rho.$$

Changing the order gives (P), with the nonnegative kernel $[|y|^{2-n} - r^{2-n}]/[n(n-2)\omega_n]$ when $n \geq 3$. Replace $S_u(0, r)$ by the boundary average of g. The kernel's singularity is locally integrable, justifying the change for bounded f; signed f is treated by absolute integrability. □

4. A direct maximum-principle proof.

Solution. For $\varepsilon > 0$, $v = u + \varepsilon|x|^2$ satisfies $\Delta v = 2n\varepsilon > 0$. At an interior maximum its Hessian would be negative semidefinite and $\Delta v \leq 0$, impossible. Thus v reaches its maximum on the boundary. Bound the added quadratic uniformly on bounded U and let $\varepsilon \downarrow 0$. □

传播速度来自能量通量

球半径以速度 1 收缩时，移边界损失足以压住波的能量流入。热核虽处处为正，波的锥外影响却严格为零；这是两个方程的本质区别。

5. Subharmonic functions. Show the mean inequality and boundary maximum principle for $\Delta v \geq 0$; prove that $\phi(u)$ and $|Du|^2$ are subharmonic for harmonic u and smooth convex ϕ.

Solution. Formula (M) makes spherical means nondecreasing, and (SB) gives $v(x) \leq A_v(x, r)$. For the boundary maximum argument use $v + \varepsilon|x|^2$, as in Problem 4. The chain rule gives $\Delta\phi(u) = \phi''(u)|Du|^2 + \phi'(u)\Delta u \geq 0$. Since harmonic functions are smooth, differentiating the PDE is valid:

$$\Delta|Du|^2 = 2|D^2u|^2 + 2Du \cdot D\Delta u = 2|D^2u|^2 \geq 0.$$

□

凸性与平均相容

调和函数经凸函数变换后不一定仍调和，却成为次调和。Jensen 不等式给直觉，链式法则中的 $\phi''(u)|Du|^2$ 给出可核验的原因。

6. A uniform Poisson bound.

Solution. Enclose U in $B_R(0)$, put $F = \|f\|_\infty$, $G = \|g\|_\infty$, and set $b(x) = G + F(R^2 - |x|^2)/(2n)$. Then $b \geq G$ on ∂U and $-\Delta b = F$. The function $u - b$ is subharmonic because $-\Delta(u - b) = f - F \leq 0$; its boundary values are nonpositive. Problem 5 yields $u \leq b$. Repeat for $-u$, giving $\|u\|_\infty \leq G + R^2F/(2n)$. □

7. Explicit Harnack bounds on a ball.

Solution. For positive harmonic u on B_R with continuous boundary values, use (PB), $R - |x| \leq |x - y| \leq R + |x|$, and $\int_{\partial B_R} u = n\omega_n R^{n-1} u(0)$:

$$R^{n-2}\frac{R - |x|}{(R + |x|)^{n-1}}u(0) \leq u(x) \leq R^{n-2}\frac{R + |x|}{(R - |x|)^{n-1}}u(0).$$

If u is only given on the open ball, first use radii $\rho < R$ with $|x| < \rho$, then let $\rho \uparrow R$. □

8. Verify the ball Poisson formula.

Solution. The full verification following (PB) establishes mass one using the known solution 1, harmonicity by direct differentiation, and boundary convergence by a near/far split. Positivity controls the near error, and the factor $R^2 - |x|^2$ kills the far error. This proves existence; the maximum principle supplies uniqueness. □

2.5.2 Problems 9–17: boundaries and the heat equation

9. Lipschitz boundary data can give an unbounded normal derivative. Let bounded continuous half-space data satisfy $g(z) = |z|$ for $|z| \le 1$, $n \ge 2$. Show the Poisson solution has unbounded gradient near zero.

Solution. The boundary value at zero is zero. For $\lambda > 0$,

$$\frac{u(\lambda e_n)}{\lambda} = \frac{2}{n\omega_n} \int_{\mathbb{R}^{n-1}} \frac{g(z)}{(|z|^2 + \lambda^2)^{n/2}}\, dz.$$

The portion $|z| > 1$ is uniformly bounded in absolute value, by boundedness of g and the integrable radial tail. On $\lambda < |z| < 1$, polar coordinates bound the remaining positive integral below by $c_n \int_\lambda^1 d\rho/\rho$, diverging. A bounded gradient would bound this difference quotient, a contradiction. □

10. Odd reflection across a flat zero boundary.

Solution. First assume u is C^2 up to the flat boundary of a half-ball, harmonic inside, and zero on the flat portion. Tangential first and second derivatives vanish there. The PDE then makes $u_{nn} = 0$ there. For the odd extension, tangential derivatives change sign below the plane, normal first derivatives match, mixed derivatives match, and normal second derivatives match because their boundary values are zero. Thus the extension is C^2 and harmonic.

For only continuous boundary regularity, take a small full ball centered on the flat plane. Prescribe its continuous odd-reflected spherical data and solve by (PB). Uniqueness makes this harmonic extension odd, hence zero on the plane. On the upper half-ball it and u have the same full boundary data, so the maximum principle makes them equal. Its lower half is the desired odd reflection. Such balls cover the interface. □

反射要检查到二阶

奇延拓的函数值匹配只是第一步。要直接证明 C^2，还需用零边界值消去切向二阶导数，再用方程推出法向二阶导数为零。

11. Kelvin transform. Show $\overline{u}(x) = |x|^{2-n} u(x/|x|^2)$ is harmonic when u is harmonic, on the inverted domain away from zero.

Solution. Put $r = |x|$, $z = x/r^2$ and $e = x/r$. Then $Dz = r^{-2}(I - 2ee^T)$, so $Dz(Dz)^T = r^{-4}I$, and $\Delta z_i = -2(n-2)x_i/r^4$. Also $\Delta r^{2-n} = 0$ off zero. In the Laplacian product and chain rule, the first-derivative terms cancel: the contribution from $r^{2-n}\Delta z_i$ is opposite to that from $2D(r^{2-n})\cdot Dz_i$. The remaining Hessian term is

$$\Delta_x\big[r^{2-n}u(z)\big] = r^{-n-2}(\Delta u)(z).$$

This proves the claim, including $n = 2$ with prefactor one. □

12. Parabolic scaling and its infinitesimal generator.

Solution. For $\lambda > 0$, define $u_\lambda(x,t) = u(\lambda x, \lambda^2 t)$. The chain rule gives

$$Lu_\lambda = \lambda^2 (Lu)(\lambda x, \lambda^2 t) = 0.$$

Differentiate this identity in λ at one. The result is $L(x \cdot Du + 2tu_t) = 0$. Nonzero negative λ also gives a solution by spatial reflection; $\lambda = 0$ requires data at time zero and is not needed for the argument. □

13. A self-similar profile. For $u(x,t) = v(x/\sqrt{t})$ in one space dimension, derive the kernel.

Solution. For $z = x/\sqrt{t}$, $u_t = -zv'/(2t)$ and $u_{xx} = v''/t$. Thus $v'' + (z/2)v' = 0$, so $v' = ce^{-z^2/4}$ and $v(z) = c\int_0^z e^{-q^2/4}\,dq + d$. Choose $c = (4\pi)^{-1/2}$ and $d = 1/2$ to obtain initial step data. Differentiating in x gives $u_x = (4\pi t)^{-1/2} e^{-x^2/(4t)} = K(x,t)$. The distributional derivative of the initial step is δ_0, so this construction indeed produces the fundamental solution. □

14. Heat flow with reaction and forcing.

Solution. For $u_t - \Delta u + cu = f$, let $v = e^{ct}u$. Then $Lv = e^{ct}f$ and $v(\cdot,0) = g$. Duhamel gives

$$u(x,t) = e^{-ct}(K_t * g)(x) + \int_0^t e^{-c(t-s)}(K_{t-s} * f(\cdot,s))(x)\,ds.$$

□

15. Heat equation on a half-line with boundary forcing. For zero initial data and boundary value $g(t)$, $g(0) = 0$, derive

$$u(x,t) = \int_0^t \frac{x\,e^{-x^2/[4(t-s)]}}{\sqrt{4\pi}(t-s)^{3/2}} g(s)\,ds, \qquad x > 0.$$

Solution. The zero-boundary half-line heat Green kernel is $K(x-y,\tau) - K(x+y,\tau)$. Its inward boundary flux gives the Poisson kernel $P(x,\tau) = -2K_x(x,\tau) = xe^{-x^2/(4\tau)}/(\sqrt{4\pi}\tau^{3/2})$. It satisfies the heat equation for $x, \tau > 0$. The substitution $z = x/(2\sqrt{\tau})$ shows $\int_0^\infty P(x,\tau)\,d\tau = 1$. As $x \downarrow 0$, the mass on $\tau > \delta$ tends to zero. Thus $P(x,\cdot)$ is an approximate identity at time zero. After extending g by zero to negative times, its convolution tends to $g(t)$; compatibility $g(0) = 0$ makes the corner continuous. For fixed $x > 0$, the exponential kills all singular powers at $\tau = 0$, justifying differentiation and the PDE. The time interval shrinks to zero at the initial limit, proving the zero initial datum. □

16. Direct parabolic maximum principle.

Solution. The argument $v = u - \varepsilon t$ in §2.3.3 supplies the proof. At a top-time maximum the left derivative is nonnegative; one does not need to pretend that the terminal point is a two-sided time maximum. □

17. Subsolutions and convex transformations.

Solution. For $Lv \le 0$, (HR) and its limit at zero imply $v(x,t) \le M_r[v]$. The perturbed maximum argument from Problem 16 applies to subsolutions, giving the boundary maximum principle on bounded cylinders. If ϕ is smooth and convex and $Lu = 0$, then

$$L\phi(u) = \phi'(u)Lu - \phi''(u)|Du|^2 \le 0.$$

For a smooth caloric u, spatial and time derivatives are caloric, and $L(w^2) = -2|Dw|^2$ whenever $Lw = 0$. Sum for $w = \partial_i u$ and $w = u_t$:

$$L(|Du|^2 + u_t^2) = -2|D^2u|^2 - 2|Du_t|^2 \le 0.$$

□

椭圆与热的凸性公式

椭圆侧是 $\Delta\phi(u) = \phi''(u)|Du|^2 \ge 0$；热算子侧是 $L\phi(u) = -\phi''(u)|Du|^2 \le 0$。符号不同，得到的平均不等式方向一致。

2.5.3 Problems 18–24: waves, systems, and energy

18. Stokes' rule. If u has wave data $(0,h)$, show $v = u_t$ has data $(h,0)$.

Solution. Time differentiation commutes with $\partial_t^2 - \Delta$, so v solves the same PDE. At zero, $v = h$ and $v_t = u_{tt} = \Delta u = \Delta 0 = 0$. This is also the reason the displacement part of the solution is a time derivative of the velocity solution operator. □

19. Null coordinates and one-way waves.

Solution. The equation $u_{xy} = 0$ first makes u_x independent of y; integrating gives $u = F(x) + G(y)$. For $\xi = x + t$, $\eta = x - t$, the wave operator is $-4\partial_\xi\partial_\eta$, so $u = F(x+t) + G(x-t)$. The initial conditions give $F' = (g' + h)/2$, $G' = (g' - h)/2$, producing (DA). A purely right-moving profile $G(x-t)$ requires $h = -g'$. A purely left-moving profile $F(x+t)$ requires $h = g'$. Additive constants can be allocated to either profile without changing these derivative conditions. □

20. Undistorted radial profiles. For nonzero attenuation $\alpha(r)$ and nonnegative delay $\beta(r)$, $\beta(0) = 0$, suppose $\alpha(r)\phi(t - \beta(r))$ solves the wave equation for every smooth profile ϕ.

Solution. Insert the form into $u_{tt} - u_{rr} - (n-1)u_r/r$. Independence of ϕ, ϕ', ϕ'' gives

$$\alpha(1 - \beta'^2) = 0, \qquad 2\alpha'\beta' + \alpha\beta'' + \frac{n-1}{r}\alpha\beta' = 0,$$

$$\alpha'' + \frac{n-1}{r}\alpha' = 0.$$

For a nontrivial smooth attenuation, the first and the nonnegative-delay condition give $\beta(r) = r$; the middle equation gives $\alpha = Cr^{-(n-1)/2}$. Substitution into the last equation yields $(n-1)(n-3)C = 0$. Thus $n = 1$, $\alpha = C$, or $n = 3$, $\alpha = C/r$. The identically zero attenuation is a trivial exception and has to be excluded to make the dimension claim true. □

量词“任意波形”很强

把 ϕ, ϕ', ϕ'' 的系数分别置零，依赖于要求对所有光滑 ϕ 成立。若只要求某一个特定波形，不能推出相同的维数限制。

21. Maxwell and elastic waves.

Solution. Differentiate $E_t = \operatorname{curl} B$ and use $B_t = -\operatorname{curl} E$:

$$E_{tt} = -\operatorname{curl}\operatorname{curl} E = -\nabla \operatorname{div} E + \Delta E = \Delta E.$$

Likewise $B_{tt} = \Delta B$, using the divergence constraints. For elasticity, $u_{tt} - \mu\Delta u - (\lambda + \mu)\nabla \operatorname{div} u = 0$. Taking divergence and curl respectively gives

$$w_{tt} - (\lambda + 2\mu)\Delta w = 0, \qquad \mathbf{w}_{tt} - \mu\Delta \mathbf{w} = 0,$$

where $w = \operatorname{div} u$ and $\mathbf{w} = \operatorname{curl} u$. When $\mu > 0$ and $\lambda + 2\mu > 0$, the longitudinal and transverse speeds are $\sqrt{\lambda + 2\mu}$ and $\sqrt{\mu}$. □

22. Telegraph equation from velocity switching. Given $u_t + u_x = d(v - u)$ and $v_t - v_x = d(u - v)$, show each component satisfies $w_{tt} + 2dw_t - w_{xx} = 0$.

Solution. Let $a = u+v$, $b = u-v$. Adding and subtracting gives $a_t + b_x = 0$ and $b_t + a_x = -2db$. Differentiate the first in time: $a_{tt} = -b_{xt} = a_{xx} + 2db_x = a_{xx} - 2da_t$. Differentiate the second in time and use $a_{xt} = -b_{xx}$: $b_{tt} = b_{xx} - 2db_t$. Linear combinations give the same telegraph equation for u, v. □

23. A compact space-time source and its outgoing shape. In the diamond $S = \{|x| + |t-2| < 1\}$, take $f = 1$ below $t = x+2$ and $f = -1$ above it, zero elsewhere. Solve with zero wave data and describe u for $t > 3$.

Solution. The one-dimensional Duhamel formula is

$$u(x,t) = \frac{1}{2}\int_0^t \int_{x-(t-s)}^{x+(t-s)} f(y,s)\,dy\,ds.$$

Set $\alpha = s + y$, $\beta = s - y$. Then S becomes $1 < \alpha < 3$, $1 < \beta < 3$, and $dy\,ds = \frac{1}{2}d\alpha\,d\beta$. The sign depends only on β: it is $+1$ for $\beta < 2$, -1 for $\beta > 2$. The backward cone imposes $\alpha \le t + x$, $\beta \le t - x$. Define

$$F(a) = \min\{2, \max\{0, a-1\}\},$$

$$H(b) = \begin{cases} 0, & b \le 1, \\ b-1, & 1 < b \le 2, \\ 3-b, & 2 < b < 3, \\ 0, & b \ge 3. \end{cases}$$

The integrals factor, giving $u(x,t) = \frac{1}{4}F(t+x)H(t-x)$. If $t > 3$ and $H(t-x) \neq 0$, then $t + x = 2t - (t-x) > 3$, so $F(t+x) = 2$. Therefore

$$u(x,t) = \tfrac{1}{2}H(t-x) \quad (t > 3).$$

It is a right-moving triangular pulse of height $1/2$, supported on $t - 3 < x < t - 1$, with peak at $x = t - 2$. The sign-balanced source cancels the other outgoing pulse. This piecewise forcing is interpreted through the integral formula; across its jump lines the PDE is read distributionally. □

特征坐标把菱形变方形

$\alpha = s + y$, $\beta = s - y$ 同时把波锥不等式变成坐标上界。二重积分才能分解成两个一维截断积分；漏掉 Jacobian 的 $1/2$ 会让最终脉冲高度错一倍。

24. Equipartition in one dimension. For compactly supported smooth g, h, show total wave energy is constant and kinetic and potential energies eventually coincide.

Solution. Write $u = F(x+t) + G(x-t)$ as in Problem 19. Although the primitives F, G need not be compactly supported, their derivatives $F' = (g' + h)/2$, $G' = (g' - h)/2$ are. Since $u_t = F' - G'$ and $u_x = F' + G'$,

$$k(t) + p(t) = \int_{\mathbb{R}} \left(F'(x+t)^2 + G'(x-t)^2\right) dx$$

is independent of t by translation invariance. Also

$$k(t) - p(t) = -2\int_{\mathbb{R}} F'(x+t)G'(x-t)\,dx.$$

For sufficiently large t the translated compact supports are disjoint, so this integral is zero. If the initial supports lie in $[-R, R]$, every $t > R$ suffices. □

2.6 References

The primary source is Lawrence C. Evans, *Partial Differential Equations*, second edition, Chapter 2, printed pp. 17–87 (local PDF pp. 34–104). The exposition here is reconstructed for study; source theorem numbering is retained only as an explicit source identifier.

Evans's §2.6 attributes the heat-ball theorem and the associated strong maximum principle to N. Watson, *Proceedings of the London Mathematical Society* 26 (1973), pp. 385–417. The proof above follows the source's scaling/logarithm method and adds the full mass computation, tip estimates, scale-defect identity, and moment interpretation. The latter supplements are editorial deductions; no separate Watson-paper comparison is claimed. The English source omits the normalization calculation; the Chinese translator's footnote on printed p. 37 suggests polar coordinates and a Gamma substitution. The present proof spells out those constants.

The source points to Gilbarg–Trudinger, Chapters 2–4, for elliptic theory; John and Friedman for the heat equation; Mikhailov for regularity; Payne for backward uniqueness; and Folland and Strauss for wave representation formulas. These are source reading directions, not claims that those books were independently consulted for this draft.

The Chinese chapter opening (PDF p. 31) provides section starts at printed pp. 13, 14, 30, 43, 56, and 60 for §2.1–§2.6. Its mean-value and heat-ball pages were visually checked for terminology and formulas as recorded in the chapter log. This is not a claim of a line-by-line comparison of the entire Chinese chapter.

Scope and review. Elementary calculus, measure-theoretic limit theorems, the Gaussian integral and Gamma integral, and the explicitly invoked complex-analytic parameter-integral/Cauchy facts are prerequisites. General-domain existence of harmonic boundary correctors is left to later Dirichlet theory. All four model equations, their representation methods, and the 24 source exercises have study explanations here; mathematical self-checks and numerical moment checks are not independent mathematical review.